

Integration — Lesson 8 Test: FTC Practice, Polynomial Fluency

10 questions · show all three steps in the box (antiderivative, plug in top, plug in bottom and subtract) · fractions are welcome as final answers · no calculator needed

Name: _____ Date: _____ Score: _____ / 10

1. Evaluate $\int_0^4 x \, dx$.

1 pt

2. Evaluate $\int_0^2 x^2 \, dx$. Leave your answer as a fraction.

1 pt

3. Evaluate $\int_0^5 3 \, dx$, then check your answer with a rectangle picture.

1 pt

4. Evaluate $\int_0^2 (2x + 5) \, dx$.

1 pt

5. Evaluate $\int_0^1 x^3 dx$. Leave your answer as a fraction. (Careful: divide by the NEW power.) 1 pt

6. Evaluate $\int_1^2 x^2 dx$. Leave your answer as a fraction. 1 pt

7. Evaluate $\int_1^3 (2x + 1) dx$. 1 pt

8. Evaluate $\int_0^1 (6x^2 + 2x) dx$. 1 pt

9. Evaluate $\int_1^2 (3x^2 - 2x) dx$. Show $F(1)$ even if it turns out to be simple.

1 pt

10. Evaluate $\int_0^2 (x^3 - x + 2) dx$. Before plugging in, CHECK your antiderivative by differentiating it, and write "check ✓" next to it.

1 pt
